

Assignment3 Report

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2026-05-27

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Executive summary

Introduction

Methodology

Dataset Introduction

This study uses the Australian Rental Market Data 2026 dataset obtained from Kaggle (Karunaratna 2026). The dataset provides a national snapshot of residential rental listings across Australia in 2026 and includes information on weekly rental prices, dwelling characteristics, and geographic attributes. To investigate rental price pressure across regions and dwelling types, variables related to rental price, location, dwelling category, and property characteristics were retained for analysis.

Before analysis, the dataset was reviewed for missing values, duplicate observations, and variable suitability. One duplicated observation was identified and removed to avoid repeated records. Missing values occurred in variables outside the scope of this study and therefore were not treated. Variables unrelated to the research objective, including listing text and detailed address information, were excluded to reduce unnecessary complexity. The final cleaned dataset retained variables describing rental price, geographic location, dwelling type, and property characteristics.

Dataset Description

The cleaned dataset contains 6766 observations and 9 variables. The retained variables include numeric measures describing rental price and dwelling characteristics (`price_display`, `bedrooms`, `bathrooms`, and `parking_spaces`) and categorical variables describing location and dwelling category (`state`, `locality`, `suburb`, and `propertyType`). These variables provide the basis for comparing rental conditions across Australia in 2026.

Dataset Summary

Table 1 summarises rental prices across dwelling types represented in the cleaned Australian rental dataset for 2026. Houses accounted for the largest share of observations (4,089 properties), followed by apartments (1,369 properties) and units (761 properties), indicating that the dataset primarily represents conventional residential dwellings. Median weekly rental prices differed across dwelling categories, ranging from AUD 430 for studios to AUD 970 for acreage and semi-rural properties.

Table 1: Summary statistics of rental prices by dwelling type.

propertyType	number_of_properties	avg_rent	median_rent
acreage/semi-rural	17	1094.1	970
terrace	7	871.4	750
apartment	1369	770.7	700
townhouse	142	741.3	700
house	4089	758.0	695
duplex/semi-detached	134	787.0	650
villa	25	624.8	590
unit	761	587.9	550
other	33	525.0	540
flat	85	519.8	510
studio	104	484.2	430

In addition to dwelling type characteristics, Figure 1 shows the distribution of rental properties across Australia in 2026. Most observations were from New South Wales, followed by Victoria and Western Australia, indicating broad geographic coverage of the Australian rental market for subsequent analysis.

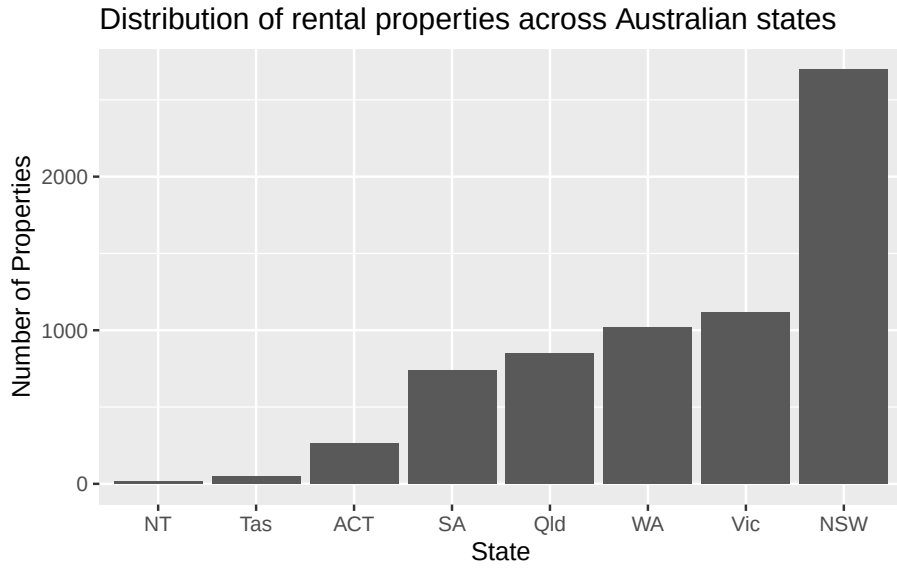


Figure 1: Distribution of rental properties across Australian states

Results

Rental stress is defined here as weekly rent exceeding 30% of the median weekly household income for each state, following the standard affordability benchmark (Australian Institute of Health and Welfare 2025). Across the full dataset, 59.5% of listings meet this threshold, indicating that the majority of available rentals in 2026 are unaffordable for a median-income household.

Figure 2 shows the proportion of listings classified as rental stress by state. Queensland recorded the highest stress rate (75.1%), followed by Western Australia (71.6%) and Tasmania (67.3%). Despite having the highest median weekly rent, the Australian Capital Territory had the lowest stress rate (11.7%), reflecting its substantially higher median household income relative to other states. New South Wales, while recording a moderate stress rate of 62.2%, had the largest absolute number of stressed listings (1,678 properties) due to its volume of listings.

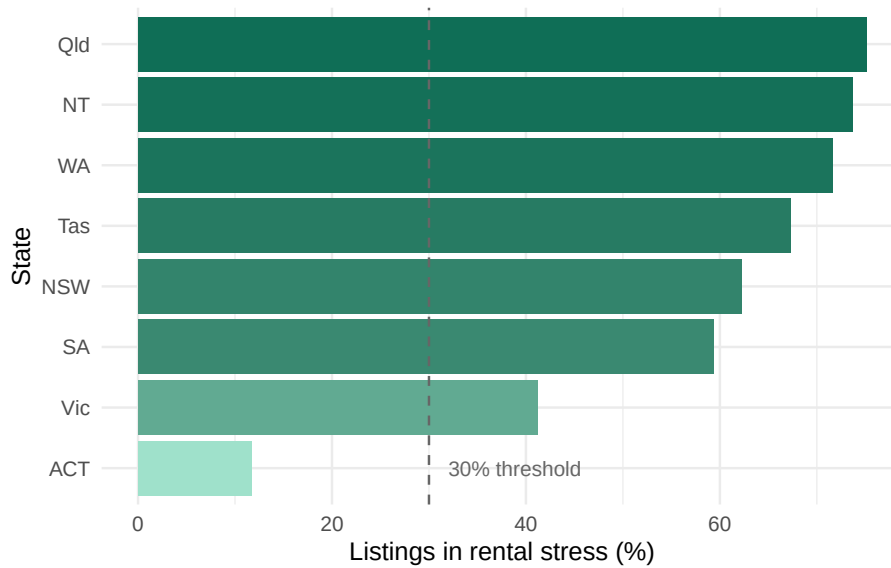


Figure 2: Proportion of rental listings exceeding the 30% income threshold (rental stress) by Australian state, 2026.

Rental stress by dwelling type

Table 2 summarises the rental stress rate and median weekly rent for each dwelling type. Acreage and semi-rural properties recorded the highest stress rate (88.2%) and the highest median rent (\$970/wk), making them the least affordable dwelling category. Houses (65.2%) and apartments (63.1%) followed,

representing the most common dwelling types in the dataset. Studios recorded the lowest stress rate (11.5%) and the lowest median rent (\$430/wk), making them the most accessible option for renters on median incomes.

Table 2: Rental stress rate and median weekly rent by dwelling type.

propertyType	Number of listings	Median rent (\$/wk)	Stress rate (%)
acreage/semi-rural	17	970	88.2
terrace	7	750	71.4
townhouse	142	700	65.5
house	4089	695	65.2
apartment	1369	700	63.1
duplex/semi-detached	134	650	51.5
villa	25	590	44.0
unit	761	550	36.5
other	33	540	21.2
flat	85	510	11.8
studio	104	430	11.5

Discussion, Conclusion and Recommendations

References

- Australian Institute of Health and Welfare. 2025. *Housing Assistance in Australia: Financial Assistance*. <https://www.aihw.gov.au/reports/housing-assistance/housing-assistance-in-australia/contents/financial-assistance>.
- Karunaratna, Kanchana. 2026. *Australian Rental Market Data 2026*. Kaggle. <https://doi.org/10.34740/KAGGLE/DS/9582256>.